Renal Physiology

Session 1

Introduction to kidney functions, renal organization, and the basic mechanisms of urine formation

Reference:

Guyton and Hall Textbook of Medical Physiology (edition 2026), Chapter 26.

Part 1. Why do we need kidneys?

Before we discuss the structure of the kidney, I want you to consider a simple question.

What would happen if the kidneys suddenly stopped working?

Actually, the situation is much more serious. The kidneys do much more than produce urine. They control the amount of water in the body, regulate important electrolytes, help control blood pressure, remove metabolic waste products, and maintain the chemical environment required for cells to function.

Therefore, our first principle is:

The primary physiological role of the kidneys is to maintain homeostasis by regulating the composition and volume of body fluids and by removing waste products.

What is homeostasis?

Homeostasis means maintaining a relatively stable internal environment despite changes inside and outside the body.

Part 2. The major functions of the kidneys

I recommend introducing the functions in six groups.

2.1 Excretion of metabolic waste products

The body continuously produces waste products through normal metabolism.

Important examples include:

- Urea: produced mainly in the liver from the metabolism of amino acids.

- Creatinine: produced from creatine metabolism, especially in skeletal muscle.
- Uric acid: produced during the breakdown of purines.
- Metabolites of hormones and other compounds: some are eliminated through renal excretion.
- Drugs and their metabolites: many are eliminated partly or mainly by the kidneys.

Blood carries these substances to the kidneys. Some are filtered from the blood into the renal tubules. Depending on the substance, the tubules may reabsorb some of it or secrete additional amounts into the tubular fluid. The final urine carries the excreted substances out of the body.

A useful distinction for students is that filtration is not the same as excretion. A substance can be filtered and then reabsorbed, so it does not all appear in the urine.

Clinical correlation: creatinine

Serum creatinine is commonly used to assess kidney function. When the glomerular filtration rate (GFR) decreases, serum creatinine usually increases.

However, creatinine is not a perfect marker. Its concentration is also affected by muscle mass, age, diet, and some medications. A patient with very low muscle mass may have significant kidney dysfunction despite a creatinine value that does not appear markedly elevated.

2.2 Regulation of water balance

Water is essential for:

- maintaining blood volume,
- transporting nutrients,
- supporting cellular chemical reactions,
- controlling body temperature,
- maintaining appropriate extracellular fluid osmolality.

The kidneys regulate how much water leaves the body in urine.

Example: dehydration

Imagine that a patient has been working outside in hot weather and has lost a large amount of water through sweating.

The body detects that water is becoming insufficient relative to solutes. The release of antidiuretic hormone increases. This hormone acts on the kidney's collecting ducts, allowing more water to be reabsorbed into the blood.

As a result, the urine volume decreases and the urine becomes more concentrated.

Example: drinking excess water

When a person drinks a large amount of water, plasma osmolality tends to decrease. Antidiuretic hormone secretion decreases, the collecting ducts become less permeable to water, and more water is excreted in the urine.

Key principle: The kidneys can conserve water or excrete excess water according to the body's needs.

2.3 Regulation of electrolytes

Electrolytes are charged particles dissolved in body fluids.

Important examples include:

- sodium (Na^+),
- potassium (K^+),
- chloride (Cl^-),
- calcium (Ca^{2+}),
- phosphate,
- bicarbonate (HCO_3^-).

These substances are essential for nerve conduction, muscle contraction, cellular function, and fluid balance.

Sodium

Sodium is the major cation of extracellular fluid. The total amount of sodium in the body is a major determinant of extracellular fluid volume.

When the kidneys retain more sodium, water often follows, increasing extracellular fluid volume. When they excrete more sodium, extracellular fluid volume tends to decrease.

With a sudden 10-fold increase in sodium intake from a low level of 30 mEq/day to a high level of 300 mEq/day. Within 2 to 3 days after raising the sodium intake, renal excretion also increases to about 300 mEq/day so that the balance between intake and output is rapidly reestablished. However, during the 2 to 3 days of renal adaptation to the high sodium intake, there is usually a modest accumulation of sodium that raises extracellular fluid volume slightly and triggers hormonal changes and other compensatory responses that signal the kidneys to increase their sodium excretion.

Potassium

Potassium is the major cation inside cells. Even relatively small changes in extracellular potassium concentration can affect the electrical activity of nerves and muscles. The kidneys are the principal long-term regulators of potassium balance.

Clinical correlation: hyperkalemia

If kidney function deteriorates, potassium excretion may become inadequate. Hyperkalemia can alter cardiac electrical activity and cause dangerous arrhythmias.

This is why measuring serum potassium is particularly important in patients with acute kidney injury or advanced chronic kidney disease.

2.4 Regulation of acid–base balance

The body continuously produces acids through metabolism.

The kidneys help maintain blood pH by:

1. Reabsorbing filtered bicarbonate.
2. Secreting hydrogen ions into the tubular fluid.
3. Generating new bicarbonate and adding it to the blood.
4. Excreting acid, especially as ammonium and titratable acid.

The lungs also regulate acid–base balance by controlling carbon dioxide elimination.

Explain the difference:

- Lungs: can change carbon dioxide elimination rapidly.
- Kidneys: regulate bicarbonate and net acid excretion, with important adjustments developing over hours to days.

Clinical correlation: metabolic acidosis

In significant renal dysfunction, the kidneys may not excrete enough acid or generate enough new bicarbonate. This can produce metabolic acidosis.

Metabolic acidosis may also occur in severe diarrhea, diabetic ketoacidosis, and other conditions. Therefore, not every case of metabolic acidosis is caused by kidney disease.

2.5 Regulation of arterial blood pressure

The kidneys regulate arterial pressure, especially over the long term, by controlling sodium balance, water balance, and extracellular fluid volume.

They also release renin, an enzyme that initiates the renin–angiotensin–aldosterone system (RAAS). The basic sequence is:

Reduced effective renal perfusion or other appropriate stimuli → renin release → angiotensin II formation → vasoconstriction and stimulation of aldosterone secretion → increased renal sodium retention under appropriate conditions → support of extracellular fluid volume and arterial pressure.

The exact response depends on the patient's volume status, renal function, and other regulatory signals.

Clinical correlation

Chronic kidney disease can contribute to hypertension through sodium retention, volume expansion, and altered hormonal regulation. In the opposite direction, severe hypotension can reduce renal perfusion and cause acute kidney injury.

This illustrates an important relationship:

The kidneys help regulate blood pressure, while blood pressure and renal perfusion also affect kidney function.

2.6 Endocrine functions of the kidneys

The kidneys also produce hormones or activate hormone precursors.

Substance	Main physiological function
Renin	Initiates the RAAS, helping regulate blood pressure and sodium balance
Erythropoietin (EPO)	Stimulates red blood cell production in the bone marrow
Active vitamin D (calcitriol)	Increases intestinal calcium and phosphate absorption and supports bone mineral balance

The kidneys also produce locally acting substances that influence renal blood flow and tubular function.

Clinical correlation: anemia in chronic kidney disease

When functioning renal tissue is lost, erythropoietin production may decrease. The bone marrow then receives less stimulation to produce red blood cells, contributing to anemia.

Clinical correlation: bone and mineral disorders

The kidneys convert 25-hydroxyvitamin D into its active form, 1,25-dihydroxyvitamin D3 (calcitriol). Advanced kidney disease can reduce calcitriol production and disturb calcium,

phosphate, and parathyroid hormone regulation, contributing to chronic kidney disease–mineral and bone disorder.

2.7 Glucose Synthesis.

The kidneys synthesize glucose from amino acids and other precursors during prolonged fasting, a process referred to as gluconeogenesis. The kidneys' capacity to add glucose to the blood during prolonged periods of fasting rivals that of the liver.

Transition: "Now that we understand why the kidneys are important, let us examine how their structure allows them to perform these functions."

Part 3. Functional anatomy of the kidneys

3.1 Where are the kidneys located?

The kidneys are two bean-shaped organs located on the posterior abdominal wall, outside the peritoneal cavity. This location is called retroperitoneal. Each kidney typically extends from approximately T12 to L3.

Each adult kidney weighs approximately 150 grams and is about the size of a clenched fist.

The medial border contains a region called the hilum. The renal artery enters through this region, while the renal vein and ureter leave the kidney through it. The kidney is surrounded by a tough fibrous capsule that protects its delicate inner structures.

The urinary tract includes:

- Kidneys — form urine.
- Ureters — transport urine to the bladder.
- Urinary bladder — stores urine.
- Urethra — carries urine out of the body.

Clinical correlation: urinary obstruction

"A kidney can produce urine normally, but if a ureter becomes obstructed by a stone, urine cannot drain normally. Pressure may increase upstream, causing dilation of the urinary tract and, if obstruction persists, damage to renal tissue."

This is an important distinction: urine production and urine drainage are different processes.

3.2 The cortex and medulla

A. Renal cortex

The cortex is the outer region. It contains:

- renal corpuscles, including glomeruli and Bowman's capsules;
- proximal convoluted tubules;
- distal convoluted tubules;
- portions of the collecting system.

Main physiological point: The renal cortex is an important site of filtration and tubular processing.

B. Renal medulla

The medulla is the inner region. It contains:

- loops of Henle;
- collecting ducts;
- specialized blood vessels called the vasa recta, particularly around juxtamedullary nephrons.

The medulla is organized into cone-shaped structures called renal pyramids. The tip of each pyramid is the renal papilla. The medulla is divided into 8 to 10 cone-shaped masses of tissue called renal pyramids.

Urine flows from collecting ducts through the papilla into a minor calyx, then into a major calyx, the renal pelvis, and finally the ureter. The walls of the calyces, pelvis, and ureter contain contractile elements, including smooth muscle, that propel the urine toward the bladder, where urine is stored until it is emptied by micturition.

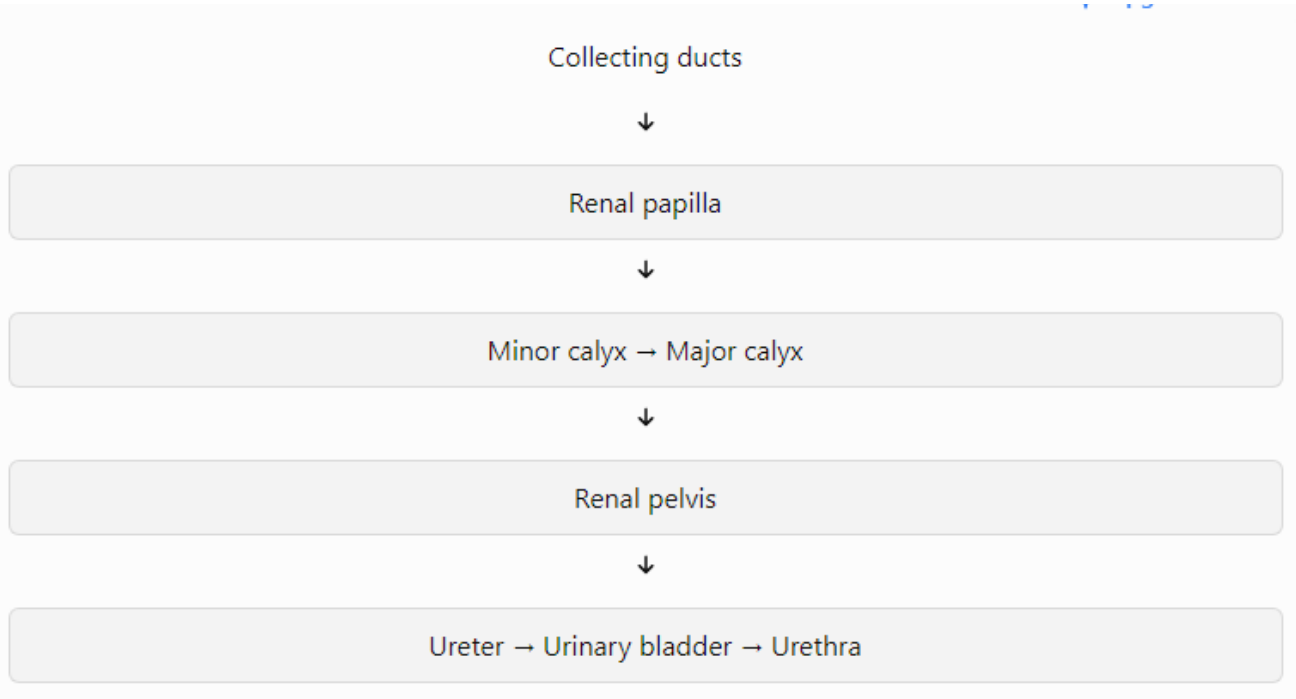
Why is the medulla important?

The medulla has a high interstitial osmolality that helps the kidneys conserve water and produce concentrated urine. This depends on the loop of Henle, the collecting ducts, urea handling, and the specialized medullary blood supply.

We will study this mechanism in detail in a later session.

3.3 Follow the path of urine

Ask students to arrange these structures in the correct order:



Part 4. Renal blood supply: why do the kidneys receive so much blood?

One of the most interesting facts about the kidneys is that they receive a large proportion of the blood pumped by the heart.

In a typical adult at rest, renal blood flow is approximately 1.1 to 1.2 liters per minute, or roughly 20–25% of cardiac output, although the exact value varies with the individual and physiological conditions.

Why do the kidneys need so much blood?

Their primary purpose is not simply to supply oxygen to renal tissue. They must also continuously examine the plasma and regulate the amounts of water and dissolved substances that remain in the body.

Renal artery

↓

Segmental arteries

↓

Interlobar arteries

↓

Arcuate arteries

↓

Interlobular arteries, also called cortical radial arteries

↓

Afferent arteriole

↓

Glomerular capillaries

↓

Efferent arteriole

↓

Peritubular capillaries and, in juxtamedullary nephrons, the vasa recta

↓

Renal veins

↓

Renal vein leaves the kidney.

The most important feature is that the glomerular capillaries lie between two arterioles.

- The afferent arteriole brings blood into the glomerulus.
- The efferent arteriole carries blood away from it.

This arrangement helps regulate glomerular capillary pressure and therefore influences filtration.

Why are there two capillary networks?

The first network, the glomerular capillaries, filters fluid from blood into Bowman's space.

The second network, the peritubular capillaries, surrounds the renal tubules and supports the exchange of water and solutes between tubular fluid and blood.

In juxtamedullary nephrons, the vasa recta also help preserve the medullary osmotic gradient needed for concentrated urine.

Clinical correlation: dehydration and shock

When blood volume or arterial pressure falls severely, renal perfusion can decrease. The body attempts to preserve circulation to vital organs, but prolonged or severe renal hypoperfusion can impair filtration and cause acute kidney injury.

This explains why a patient with severe vomiting, diarrhea, hemorrhage, or shock may develop a rising serum creatinine concentration.

However, renal blood flow and GFR are related but not identical: changes in arteriolar resistance can alter GFR even when the change in total renal blood flow is relatively small.

Part 5. The nephron — the functional unit of the kidney

A nephron is the basic functional unit of the kidney. It filters plasma and then modifies the filtered fluid to produce urine. At birth, humans have a total of about two million nephrons in their kidneys, followed by two million Bowman's capsules and glomeruli. And after birth, no new nephrons are produced. It takes about two to three years for nephrons to mature. After the age of 40, approximately 1% of nephrons are lost per year. Therefore, in an 80-year-old person, the number of nephrons is reduced to 40%.

Each kidney contains approximately one million nephrons on average, although the number varies considerably between individuals. Each nephron has two major components:

1. A renal corpuscle, where filtration begins.
2. A renal tubule, where the filtered fluid is modified."

5.1 Renal corpuscle

The renal corpuscle consists of two structures:

- Glomerulus
- Bowman's capsule

The glomerulus is a tuft of capillaries. Blood pressure drives water and many small dissolved substances across the filtration barrier into Bowman's space.

Blood cells and most plasma proteins are normally retained in the circulation.

The resulting fluid is called glomerular filtrate.

What determines which substances enter the filtrate?

Filtration depends on several factors:

- the size and properties of the substance;
- the structure and integrity of the filtration barrier;
- the forces driving filtration.

Water, sodium, glucose, urea, and many other small solutes can normally enter the filtrate. Blood cells and most albumin remain in the blood.

Clinical correlation: proteinuria

If the glomerular filtration barrier is damaged, abnormal amounts of protein can enter the urine.

Persistent albuminuria may indicate kidney damage, including damage associated with diabetes or hypertension.

A large amount of protein in urine is therefore not simply a change in urine composition; it can be a clue to disease of the renal filtration barrier.

5.2 Proximal tubule

The proximal tubule receives the newly formed filtrate.

Its major function is bulk reabsorption.

Under normal conditions, it reabsorbs approximately 65–70% of the filtered sodium and water, most filtered bicarbonate, and essentially all filtered glucose and amino acids when the filtered loads remain within normal reabsorptive capacity.

These are approximate physiological values, not fixed amounts under every condition.

Why is bulk reabsorption necessary?

Because the kidneys filter a very large volume of plasma-derived fluid every day. If most of this fluid were excreted, the body would rapidly lose water and essential substances.

The proximal tubule therefore returns most filtered water and useful solutes to the blood.

5.3 Loop of Henle

The loop of Henle descends from the cortex into the medulla and then returns toward the cortex.

It has descending and ascending limbs with different transport properties.

At an introductory level, students should remember:

- The descending limb is relatively permeable to water.
- The thick ascending limb reabsorbs sodium, potassium, and chloride but is relatively impermeable to water.
- These differences help establish the medullary osmotic gradient.

Important: The loop of Henle is not simply a tube for transporting urine. It helps create the conditions that allow the kidney to conserve water.

5.4 Distal tubule

The distal tubule further adjusts the composition of tubular fluid.

It participates in the regulation of sodium, potassium, calcium, and acid–base balance.

Its function is influenced by hormones and by signals from the nephron's own regulatory apparatus.

5.5 Collecting ducts

The collecting ducts receive fluid from multiple nephrons and provide an important final site for regulation of urine composition.

Their water permeability changes in response to antidiuretic hormone (ADH), also called vasopressin.

- When ADH is high, collecting-duct water permeability increases, allowing more water to return to the blood.
- When ADH is low, water permeability decreases, allowing more water to remain in the tubular fluid and be excreted.

Aldosterone also influences sodium reabsorption and potassium secretion in the late distal nephron and collecting system.

Clinical correlation: diabetes insipidus

If ADH production or release is deficient, or the kidneys do not respond properly to ADH, the patient may produce large volumes of dilute urine and develop excessive thirst.

This is a useful clinical example of why collecting-duct water permeability is essential for normal fluid balance.

5.6 Two types of nephrons

We have different types of nephrons, which have different sizes, structures, and functions. Two groups of them have been distinguished for us. Depending on which part of the cortex the glomeruli of these nephrons are located, these two groups are distinguished:

1. makes up 70-80% of the nephrons. These nephrons have a short loop of Henle. These nephrons cannot produce urine with high osmolality and their maximum urine osmolality is 600 mOsmol/L. Their renal corpuscle is located in the more superficial parts of the cortex. And they are called **superficial cortical nephrons**.

2. The second group is the juxtamedullary nephrons, whose glomeruli are located near the medulla but in the deeper parts of the cortex and adjacent to the corticomedullar border. These nephrons constitute 20-30% of the renal nephrons. They have a unique peritubular capillaries called **vasarecta**. In addition, these nephrons are long in the loop of Henle and play a role in concentrating urine, and the maximum osmolality of the urine produced by them is 1200-1400 mOsmol/L.

Feature	Cortical nephrons	Juxtamedullary nephrons
Glomerulus	More superficial in the cortex	Deep in the cortex, near the medulla
Loop of Henle	Usually shorter	Usually longer
Associated vessels	Mainly peritubular capillaries	Specialized vasa recta
Major teaching point	Important in filtration and tubular processing	Particularly important for producing concentrated urine

Juxtamedullary nephrons make up approximately 20–30% of nephrons in the textbook description and have long loops that extend deeply into the medulla.

Teaching emphasis: The different nephron structures allow the kidney to perform both high-volume filtration and precise regulation of final urine composition.

Part 6. The three basic processes of urine formation

This is the central concept of the first renal physiology session.

Excretion = Filtration - Reabsorption + Secretion

For a substance measured as an amount excreted per unit time, this equation describes its renal handling.